

1.1

Implement Trapezoidal rule of calculating the definite integral $\int_a^b \frac{4}{1+x^2} dx$. Your program should use MPI_Reduce to rewrite the collective communication part.

INPUT: b – the upper bound of the interval;

a – the lower bound of the interval;

n – the number of subintervals (trapezoids).

OUTPUT: The value of $\int_a^b \frac{4}{1+x^2} dx$.

1.2

Use MPI to implement the histogram program discussed in Chapter 2. Have process 0 read in the input data and distribute it among the processes. Also have process 0 printout the program. Note that the measurements are randomly generated based on *data_count*, *a*, and *b* input by the user.

INPUT: *data_count* – the number of measurements;

b – the upper bound of measurements;

a – the lower bound of measurements;

n – the number of bins.

OUTPUT: The measurements;

The range of each bin;

The number of measurements in each bin.

Note: All programs have to be presented to the instructor on the class of due date.